

Seoyeong Hwang

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About

Bridging physics, signals, and intelligence — building AI that understands and predicts complex real-world systems.

Seoyeong Hwang is a master's researcher in Artificial Intelligence at Ewha Womans University, specializing in **geometric deep learning, spatio-temporal modeling, and AI for Science**. Her research spans graph neural networks (GNNs), physics-informed learning (PINNs), and the application of these methods to large-scale physical and human systems — from biomechanical deformation prediction to nation-scale EV charging demand forecasting. Beyond academic research, she has shipped production AI systems across three startups: a **Spatio-Temporal GNN benchmark for EV demand forecasting** at Empact Labs, an **IRT-based adaptive learning recommender** at MathLab-AI (KAITLab), and a **7-agent multi-stage LLM tutoring system** as Founding AI Engineer at VStdY (Stealth). She is interested in building efficient, interpretable, and deployable AI systems where scientific rigor meets engineering execution.

Research Interests

- **Spatio-Temporal Graph Neural Networks**, Geometric Deep Learning, Physics-Informed Learning
- **LLM Agents & Adaptive Learning**: Item Response Theory, multi-agent orchestration, RAG, fine-tuning
- **Recommendation Systems**, Reinforcement Learning (Bandit Algorithms)
- **AI for Science**: Biomechanics, Human Motion Analysis, large-scale time-series infrastructure modeling

Education

MS	Ewha Womans University, Artificial Intelligence & Software	Mar 2024 – Feb 2026
	• GPA: 4.02/4.0 • Advisor: Prof. Junhyug Noh	
BS	Ewha Womans University, Chemistry & Nano Science, Brain & Cognitive Sciences (Double Major)	Mar 2018 – Feb 2024
	• GPA: 3.63/4.3 • Coursework: Computer Architecture, Comparison of Learning Algorithms, Computational Theory	

Experience

VStdY (Stealth Startup), Founding AI Engineer	Seoul, Korea Apr 2026 – Present
• Shipped an end-to-end production system as sole AI/full-stack engineer — SwiftUI iOS app on TestFlight (build 1.0(2)), FastAPI backend on Fly.io (vstdy-api.fly.dev), and Next.js 16 marketing site on Vercel (vstdy.com). • Architected a 7-agent LLM tutoring orchestrator (Einstein routes Warren/Sun Tzu/Sherlock/Newton/Director/DaVinci) on LangChain/LangSmith with per-topic IRT-style theta tracking; hardened agents with regex self-correction, language auto-detection (KR/JP/CN/ES/FR), and idempotent retry logic. • Built an LLM-driven Manim video pipeline — a CLI converts a problem statement into a narrated 3Blue1Brown-style explainer video. A layout-aware critic and persistence guard cut per-render cost from \$0.095 to \$0.055 (−42%) and latency from 85s to 50s (−41%). • Owned the full product surface — Scan & Solve flow (VNDocumentCamera → Vision LLM → step solver → OpenAI TTS), SwiftUI-faithful Figma design system (25 screens, 28 tokens, 35 Code-Connect mappings), and the production deploy chain (Docker / Fly.io / Vercel / GoDaddy DNS).	
Empact Labs Co., Ltd., AI Research Intern	Seoul, Korea Jun 2025 – Present
• Designed and benchmarked an 11-model Spatio-Temporal GNN suite (DCRNN, STGCN, MTGNN, Graph-WaveNet, D2STGNN, CaST, AGCRN, iTransformer, PatchTST, TimesNet, TimeXer) for 48-hour multi-horizon EV charging demand forecasting from 168-hour input windows.	

- Constructed a graph of 3,015 Jeju charging stations as nodes with **Haversine-weighted edges**, processed 1.7M+ status-history records into 7-channel features (kWh, occupancy, arrivals, sinusoidal hour/day encodings).
- Best models achieved **MAE 6.86 (kWh)** and **0.022 (occupancy)** with DCRNN, and **0.99 (arrivals)** with D2STGNN; results contribute to a national EV infrastructure optimization project.
- Built rolling cross-validation infrastructure, tmux-based parallel training across 33 jobs, and a self-contained benchmark library now used as the team's internal forecasting toolkit.

MathLab-AI, LLC (KAITLab), AI Engineer (Contract)

Florida, USA (Remote)
Jan 2026 – Mar 2026

- Diagnosed and analyzed the AI-driven recommendation system in the KAIT learning platform, focusing on question selection and subtopic-level personalization.
- Investigated an **Item Response Theory (IRT)**-based recommendation pipeline and its integration with backend services for adaptive learning.
- Applied **multi-armed bandit**-inspired strategies to improve dynamic question recommendation and learning efficiency.
- Reviewed legacy backend architecture — including the subtopic picker and recommendation logic — to identify system inconsistencies and performance bottlenecks.
- Conducted database-level and system-level analysis to trace recommendation behavior and improve interpretability of AI-driven learning processes.

Human Performance Lab Co., Ltd., AI Research Intern

Seoul, Korea
Jan 2025 – Feb 2026

- Led AI research for predicting 3D foot deformation under contact pressure using **Physics-Informed Graph Neural Networks**.
- Developed a **MeshGraphNet**-based architecture integrating physics constraints to improve biomechanical plausibility and accuracy.
- Achieved an **87.2% reduction in prediction error** (MAE: 3.90 → 0.50) and presented results at FBS 2025 and ISB 2025.

KIST, Bionics Research Center (BC3Lab), Research Intern

Seoul, Korea
Dec 2022 – Jul 2023

- Developed an ultrasound-based 3D vascular reconstruction pipeline using deep learning and **Res-UNet** architectures.
- Processed biosignal and neuroimaging datasets for disease classification using custom Python data pipelines.
- Improved 3D reconstruction performance through architectural optimization and data augmentation, increasing generalization across patients.

Ewha Brain Institute (EBI), Undergraduate Research Assistant

Seoul, Korea
Jul 2022

- Participated in experimental neuroscience training programs focusing on cognitive and physiological measurements.
- Conducted MRI scanning sessions and practiced neurophysiological signal acquisition, including Skin Conductance Response (SCR) recording and analysis.
- Trained in advanced neuroengineering techniques such as HBOT, NIRS, tDCS, and TMS.
- Presented a study titled “Protocol Analysis of Cognitive Experimental Process Using MRI Scan.”

Publications

Conference Proceedings

[C3] **LYRICS: Line-by-Line Lyric Generation with Joint Control of Syllables, Context, and Rhyme**

ICTC 2025

*Seoyeong Hwang**, Seoyeon Ko*, Hyunseo Kim*, Jihyun Yu, Jeonghyun Kim, Hyunsoo Cho, Junhyug Noh

[C2] **Enhancing foot deformation prediction using Graph Neural Networks**

Footwear Science 2025 (Oral)

Seoyeong Hwang, Haeun Yum, Hyesu Son, Taeyong Lee, Junhyug Noh

[C1] **Leveraging Graph Neural Networks for Accurate Foot Deformation Prediction**

ISB 2025 (Oral)

Seoyeong Hwang, Haeun Yum, Hyesu Son, Taeyong Lee, Junhyug Noh

Journal & Preprint

[J1] Towards Robust Gait Identification: A Frequency Domain Approach in Varied Surveillance Environments

IEEE Access 2026

Luca Quagliato, Yejin Song, Sewon Jang, Eugene Chung, Seoyeon Ko, *Seoyeong Hwang*, Junhyug Noh, Taeyong Lee

Patents

- **Foot Deformation Prediction Device and Method** (KR P2025-0174KR, App. No. 10-2025-0150087, Pub. 2025-10-16)
- **Method and Apparatus for Generating Lyrics Considering Rhyme** (filed 2025)

Projects

Industry R&D

AI-Driven Health Monitoring for Companion Animals using IoT Devices,
BMSMILE Co., Ltd.

Nov 2024 – Oct 2025

- Developed ML algorithms to detect early signs of illness in pets from multimodal IoT data (smart feeders, wearables).
- Built an anomaly detection pipeline combining image data and biosignals for continuous health and behavior monitoring.

National R&D

Diagnostic and Assistive Footwear Technology for People with Disabilities,
Korea Medical Device Development Fund (KMDF)

Mar 2024 – Dec 2025

- AI-development leader on a project building AI-assisted diagnostic and custom footwear technology for individuals with gait impairments.
- Designed a 3D deformation prediction framework for personalized insole design and biomechanical fitting; developed a Web API deployed alongside the iOS application.

Digital Sleeponomics: Multimodal AI System for Art-based Sleep Intervention, National Research Foundation of Korea (NRF)

Jun 2025 – Feb 2026

- AI research lead on a national interdisciplinary project integrating neuroscience, art, and AI for sleep quality and well-being.
- Developed a multimodal Transformer combining EEG, ECG, and music-based features to personalize therapeutic interventions.

Teaching Experience

TA Boostcamp AI Tech (7th), @ Upstage Co., Ltd.

Aug 2024 – Jan 2025

- Designed and authored full lecture materials for the advanced course *Model Optimization & Light-weighting*.
- Developed visual and code-based explanations of distributed training: Data Parallelism (DP), Distributed Data Parallel (DDP), and Model/Pipeline Parallelism.

TA Computer Vision, @ Ewha Womans University

Sep 2024 – Dec 2024

- Delivered practical lectures on CNN, ResNet, and ViT using PyTorch implementations.
- Guided hands-on projects in object detection and semantic segmentation focusing on efficiency and accuracy.

TA AI Applied Mathematics, @ Ewha Womans University

Mar 2024 – Jun 2024

- Assisted in teaching mathematical foundations for deep learning: linear algebra, optimization, and gradient-based learning.
- Created intuitive visual materials and practice sessions for backpropagation and numerical optimization in Python.

Lecturer Python & Scratch, @ AIT STUDIO Inc.

Dec 2023 – Jun 2024

- Taught computational thinking via Scratch to elementary students; designed Python game-development courses (Pygame) for middle/high school students.

Extracurricular Activities

- Founder** **BRAINEEWHA**, Brain & Cognitive Sciences @ Ewha Womans University Mar 2022 – Dec 2022
- Founded the “Neuro-Law” division exploring neuroscience and legal evidence via U.S. court-case neuroimaging analysis.
 - Led literature reviews on the neural mechanisms of schizophrenia.
- Member** **BCSC**, Brain and Cognitive Science Community @ Seoul National University Feb 2022 – Aug 2022
- Conducted EEG signal-processing analyses on ERP responses under visual stimulation.

Honors and Awards

- **OPIc (AL)** @ ACTFL (Certificate No. KKQY-LR2I-L1VE-YNUB-6M85)
- **Full-ride Scholarship (W4)** @ Ewha Womans University

Skills

- **Languages:** Korean (Native), English (OPIc AL)
- **Programming:** Python, Swift (SwiftUI), TypeScript, MATLAB, SQL
- **ML/DL Frameworks:** PyTorch, PyTorch Geometric, LangChain/LangSmith, Hugging Face Transformers
- **GNN & Time-Series:** DCRNN, STGCN, MTGNN, Graph-WaveNet, D2STGNN, CaST, MeshGraphNet, iTransformer, PatchTST
- **LLM Engineering:** Multi-agent orchestration, RAG, SFT/PEFT (LoRA), prompt design, function calling, OpenAI Vision & TTS, Mathpix
- **Production & DevOps:** Docker, Fly.io, Vercel, FastAPI, SQLite, Next.js, AVKit, Xcode/TestFlight, Manim, Git/GitHub, tmux, wandb
- **Design & Other:** Figma (design system, Code Connect), Blender, AutoCAD, Isaac Sim, 3D Slicer